Construction of the Hopf fibration with total space the 3-sphere

The Univalent Foundations Program

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Editorial note. Only the selected result is counted. Its local constructions and author proof are retained without generated proof completion.

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Allowed standard prerequisites (editorial).

- Univalent homotopy theory foundations. Univalence, function extensionality, dependent path transport, higher inductive pushouts/suspensions, truncations and basic pointed homotopy notions. The selected local code, fiber or gluing constructions are retained as author proof. Original numbered references identify these foundations.

Source reference. Original lemma 6.12.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1780–1780. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original theorem 7.1.11: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1793–1793. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original section 6.8: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1620–1620. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original lemma 6.4.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1878–1878. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original lemma 6.5.1: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1889–1889. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

0.1 The Hopf fibration

In this section we will define the **Hopf fibration**.

Theorem 0.1.1 (Hopf Fibration)**.** *There is a fibration H over $\mathbb{S}^2$ whose fiber over the basepoint is $\mathbb{S}^1$ and whose total space is $\mathbb{S}^3$.*

0.1.1 Fibrations over pushouts

We first start with a lemma explaining how to construct fibrations over pushouts.

Lemma 0.1.2. *Let $\mathscr{D} = (Y \xleftarrow{j} X \xrightarrow{k} Z)$ be a span and assume that we have*

- *Two fibrations $E_Y : Y \to \mathcal{U}$ and $E_Z : Z \to \mathcal{U}$.*
- *An equivalence e_X between $E_Y \circ j : X \to \mathcal{U}$ and $E_Z \circ k : X \to \mathcal{U}$, i.e.*

$$e_X : \prod_{x:X} E_Y(j(x)) \simeq E_Z(k(x)).$$

Then we can construct a fibration $E : Y \sqcup^X Z \to \mathcal{U}$ such that

- *For all $y : Y$, $E(\mathsf{inl}(y)) \equiv E_Y(y)$.*
- *For all $z : Z$, $E(\mathsf{inr}(z)) \equiv E_Z(z)$.*
- *For all $x : X$, $E(\mathsf{glue}(x)) = \mathsf{ua}(e_X(x))$ (note that both sides of the equation are paths in $\mathcal{U}$ from $E_Y(j(x))$ to $E_Z(k(x))$).*

Moreover, the total space of this fibration fits in the following pushout square:

$$\begin{array}{ccc} \sum_{(x:X)} E_Y(j(x)) & \xrightarrow{\mathsf{id}\times e_X}_{\sim} & \sum_{(x:X)} E_Z(k(x)) \xrightarrow{k\times\mathsf{id}} \sum_{(z:Z)} E_Z(z) \\ \downarrow{\scriptstyle j\times\mathsf{id}} & & \downarrow{\scriptstyle \mathsf{inr}} \\ \sum_{(y:Y)} E_Y(y) & \xrightarrow[\mathsf{inl}]{} & \sum_{(t:Y\sqcup^X Z)} E(t) \end{array}$$

Proof. We define E by the recursion principle of the pushout $Y \sqcup^X Z$. For that, we need to specify the value of E on elements of the form $\mathsf{inl}(y)$, $\mathsf{inr}(z)$ and the action of E on paths $\mathsf{glue}(x)$, so we can just choose the following values:

$$\begin{aligned} E(\mathsf{inl}(y)) &:\equiv E_Y(y), \\ E(\mathsf{inr}(z)) &:\equiv E_Z(z), \\ E(\mathsf{glue}(x)) &:= \mathsf{ua}(e_X(x)). \end{aligned}$$

To see that the total space of this fibration is a pushout, we apply the flattening lemma (original lemma 6.12.2) with the following values:

- $A :\equiv Y + Z$, $B :\equiv X$ and $f, g : B \to A$ are defined by $f(x) :\equiv \mathsf{inl}(j(x))$, $g(x) :\equiv \mathsf{inr}(k(x))$,
- the type family $C : A \to \mathcal{U}$ is defined by

$$C(\mathsf{inl}(y)) :\equiv E_Y(y) \qquad \text{and} \qquad C(\mathsf{inr}(z)) :\equiv E_Z(z),$$

- the family of equivalences $D : \prod_{(b:B)} C(f(b)) \simeq C(g(b))$ is defined to be e_X.

The base higher inductive type W in the flattening lemma is equivalent to the pushout $Y \sqcup^X Z$ and the type family $P : Y \sqcup^X Z \to \mathcal{U}$ is equivalent to the E defined above.

Thus the flattening lemma tells us that $\sum_{(t:Y \sqcup^X Z)} E(t)$ is equivalent to the higher inductive type $E^{\mathrm{tot}\prime}$ with the following generators:

- a function $\mathsf{z} : \sum_{(a:Y+Z)} C(a) \to E^{\mathrm{tot}\prime}$,
- for each $x : X$ and $t : E_Y(j(x))$, a path $\mathsf{z}(\mathsf{inl}(j(x)), t) = \mathsf{z}(\mathsf{inr}(k(x)), e_X(t))$.

Using the flattening lemma again or a direct computation, it is easy to see that $\sum_{(a:Y+Z)} C(a) \simeq \sum_{(y:Y)} E_Y(y) + \sum_{(z:Z)} E_Z(z)$, hence $E^{\mathrm{tot}\prime}$ is equivalent to the higher inductive type E^{tot} with the following generators:

- a function $\mathsf{inl} : \sum_{(y:Y)} E_Y(y) \to E^{\mathrm{tot}}$,
- a function $\mathsf{inr} : \sum_{(z:Z)} E_Z(z) \to E^{\mathrm{tot}}$,
- for each $(x, t) : \sum_{(x:X)} E_Y(j(x))$ a path $\mathsf{glue}(x, t) : \mathsf{inl}(j(x), t) = \mathsf{inr}(k(x), e_X(t))$.

Thus the total space of E is the pushout of the total spaces of E_Y and E_Z, as required. □

0.1.2 The Hopf construction

Definition 0.1.3. An **H-space** consists of

- a type A,
- a base point $e : A$,
- a binary operation $\mu : A \times A \to A$, and
- for every $a : A$, equalities $\mu(e, a) = a$ and $\mu(a, e) = a$.

Lemma 0.1.4. *Let A be a connected H-space. Then for every $a : A$, the maps $\mu(a, -) : A \to A$ and $\mu(-, a) : A \to A$ are equivalences.*

Proof. Let us prove that for every $a : A$ the map $\mu(a, -)$ is an equivalence. The other statement is symmetric. The statement that $\mu(a, -)$ is an equivalence corresponds to a type family $P : A \to \mathsf{Prop}$ and proving it corresponds to finding a section of this type family.

The type Prop is a set (original theorem 7.1.11) hence we can define a new type family $P' : \|A\|_0 \to \mathsf{Prop}$ by $P'(|a|_0) :\equiv P(a)$. But A is connected by assumption, hence $\|A\|_0$ is contractible. This implies that in order to find a section of P', it is enough to find a point in the fiber of P' over $|e|_0$. But we have $P'(|e|_0) = P(e)$ which is inhabited because $\mu(e, -)$ is equal to the identity map by definition of an H-space, hence is an equivalence.

We have proved that for every $x : \|A\|_0$ the proposition $P'(x)$ is true, hence in particular for every $a : A$ the proposition $P(a)$ is true because $P(a)$ is $P'(|a|_0)$. □

Definition 0.1.5. Let A be a connected H-space. We define a fibration over ΣA using Lemma 0.1.2.

Given that ΣA is the pushout $\mathbf{1} \sqcup^A \mathbf{1}$, we can define a fibration over ΣA by specifying

- two fibrations over $\mathbf{1}$ (i.e. two types F_1 and F_2), and

- a family $e : A \to (F_1 \simeq F_2)$ of equivalences between F_1 and F_2, one for every element of A.

We take A for F_1 and F_2, and for $a : A$ we take the equivalence $\mu(a, -)$ for $e(a)$.

According to Lemma 0.1.2, we have the following diagram:

$$\begin{array}{ccccc} A & \xleftarrow{\mathsf{pr}_2} & A \times A & \xrightarrow{\mu} & A \\ \downarrow & & \downarrow{\scriptstyle \mathsf{pr}_1} & & \downarrow \\ 1 & \longleftarrow & A & \longrightarrow & 1 \end{array}$$

and the fibration we just constructed is a fibration over ΣA whose total space is the pushout of the top line.

Moreover, with $f(x, y) :\equiv (\mu(x, y), y)$ we have the following diagram:

$$\begin{array}{ccccc} A & \xleftarrow{\mathsf{pr}_2} & A \times A & \xrightarrow{\mu} & A \\ {\scriptstyle \mathsf{id}}\downarrow & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \mathsf{id}} \\ A & \xleftarrow[\mathsf{pr}_2]{} & A \times A & \xrightarrow[\mathsf{pr}_1]{} & A \end{array}$$

The diagram commutes and the three vertical maps are equivalences, the inverse of f being the function g defined by

$$g(u, v) :\equiv (\mu(-, v)^{-1}(u), v).$$

This shows that the two lines are equivalent (hence equal) spans, so the total space of the fibration we constructed is equivalent to the pushout of the bottom line. And by definition, this latter pushout is the *join* of A with itself (see original section 6.8). We have proven:

Lemma 0.1.6. *Given a connected H-space A, there is a fibration, called the **Hopf construction**, over ΣA with fiber A and total space $A * A$.*

0.1.3 The Hopf fibration

We will first construct a structure of H-space on the circle $\mathbb{S}^1$, hence by Lemma 0.1.6 we will get a fibration over $\mathbb{S}^2$ with fiber $\mathbb{S}^1$ and total space $\mathbb{S}^1 * \mathbb{S}^1$. We will then prove that this join is equivalent to $\mathbb{S}^3$.

Lemma 0.1.7. *There is an H-space structure on the circle $\mathbb{S}^1$.*

Proof. For the base point of the H-space structure we choose base. Now we need to define the multiplication operation $\mu : \mathbb{S}^1 \times \mathbb{S}^1 \to \mathbb{S}^1$. We will define the curried form $\widetilde{\mu} : \mathbb{S}^1 \to (\mathbb{S}^1 \to \mathbb{S}^1)$ of μ by recursion on $\mathbb{S}^1$:

$$\widetilde{\mu}(\mathsf{base}) :\equiv \mathsf{id}_{\mathbb{S}^1}, \qquad \text{and} \qquad \widetilde{\mu}(\mathsf{loop}) \coloneqq \mathsf{funext}(h).$$

where $h : \prod_{(x:\mathbb{S}^1)}(x = x)$ is the function defined in original lemma 6.4.2, which has the property that $h(\mathsf{base}) :\equiv \mathsf{loop}$.

Now we just have to prove that $\mu(x, \mathsf{base}) = \mu(\mathsf{base}, x) = x$ for every $x : \mathbb{S}^1$. By definition, if $x : \mathbb{S}^1$ we have $\mu(\mathsf{base}, x) = \widetilde{\mu}(\mathsf{base})(x) = \mathsf{id}_{\mathbb{S}^1}(x) = x$. For the equality $\mu(x, \mathsf{base}) = x$ we do it by induction on $x : \mathbb{S}^1$:

- If x is base then $\mu(\mathsf{base},\mathsf{base}) = \mathsf{base}$ by definition, so we have $\mathsf{refl}_{\mathsf{base}} : \mu(\mathsf{base},\mathsf{base}) = \mathsf{base}$.
- When x varies along loop, we need to prove that

$$\mathsf{refl}_{\mathsf{base}} \cdot \mathsf{ap}_{\lambda x.\, x}(\mathsf{loop}) = \mathsf{ap}_{\lambda x.\, \mu(x,\mathsf{base})}(\mathsf{loop}) \cdot \mathsf{refl}_{\mathsf{base}}.$$

  The left-hand side is equal to loop, and for the right-hand side we have:

$$\begin{aligned}
\mathsf{ap}_{\lambda x.\, \mu(x,\mathsf{base})}(\mathsf{loop}) \cdot \mathsf{refl}_{\mathsf{base}} &= \mathsf{ap}_{\lambda x.\, (\widetilde{\mu}(x))(\mathsf{base})}(\mathsf{loop}) \\
&= \mathsf{happly}(\mathsf{ap}_{\lambda x.\, (\widetilde{\mu}(x))}(\mathsf{loop}), \mathsf{base}) \\
&= \mathsf{happly}(\mathsf{funext}(h), \mathsf{base}) \\
&= h(\mathsf{base}) \\
&= \mathsf{loop}. \qquad \square
\end{aligned}$$

Now recall from original section 6.8 that the *join* $A * B$ of types A and B is the pushout of the diagram

$$A \xleftarrow{\mathsf{pr}_1} A \times B \xrightarrow{\mathsf{pr}_2} B.$$

Lemma 0.1.8. *The operation of join is associative: if A, B and C are three types then we have an equivalence $(A * B) * C \simeq A * (B * C)$.*

Proof. We define a map $f : (A * B) * C \to A * (B * C)$ by induction. We first need to define $f \circ \mathsf{inl} : A * B \to A * (B * C)$ which will be done by induction, then $f \circ \mathsf{inr} : C \to A * (B * C)$, and then $\mathsf{ap}_f \circ \mathsf{glue} : \prod_{(t:(A*B)\times C)} f(\mathsf{inl}(\mathsf{pr}_1(t))) = f(\mathsf{inr}(\mathsf{pr}_2(t)))$ which will be done by induction on the first component of t:

$$\begin{aligned}
(f \circ \mathsf{inl})(\mathsf{inl}(a)) &:\equiv \mathsf{inl}(a), \\
(f \circ \mathsf{inl})(\mathsf{inr}(b)) &:\equiv \mathsf{inr}(\mathsf{inl}(b)), \\
\mathsf{ap}_{f\circ\mathsf{inl}}(\mathsf{glue}(a,b)) &:= \mathsf{glue}(a, \mathsf{inl}(b)), \\
f(\mathsf{inr}(c)) &:\equiv \mathsf{inr}(\mathsf{inr}(c)), \\
\mathsf{ap}_f(\mathsf{glue}(\mathsf{inl}(a), c)) &:= \mathsf{glue}(a, \mathsf{inr}(c)), \\
\mathsf{ap}_f(\mathsf{glue}(\mathsf{inr}(b), c)) &:= \mathsf{ap}_{\mathsf{inr}}(\mathsf{glue}(b,c)), \\
\mathsf{apd}_{\lambda x.\, \mathsf{ap}_f(\mathsf{glue}(x,c))}(\mathsf{glue}(a,b)) &:= \text{“}\mathsf{apd}_{\lambda x.\, \mathsf{glue}(a,x)}(\mathsf{glue}(b,c))''.
\end{aligned}$$

For the last equation, note that the right-hand side is of type

$$\mathsf{transport}^{\lambda x.\, \mathsf{inl}(a)=\mathsf{inr}(x)}(\mathsf{glue}(b,c), \mathsf{glue}(a, \mathsf{inl}(b))) = \mathsf{glue}(a, \mathsf{inr}(c))$$

whereas it is supposed to be of type

$$\mathsf{transport}^{\lambda x.\, f(\mathsf{inl}(x))=f(\mathsf{inr}(c))}(\mathsf{glue}(a,b), \mathsf{ap}_f(\mathsf{glue}(\mathsf{inl}(a), c))) = \mathsf{ap}_f(\mathsf{glue}(\mathsf{inr}(b), c)).$$

But by the previous clauses in the definition, both of these types are equivalent to the following type:

$$\mathsf{glue}(a, \mathsf{inr}(c)) = \mathsf{glue}(a, \mathsf{inl}(b)) \cdot \mathsf{ap}_{\mathsf{inr}}(\mathsf{glue}(b,c)),$$

and so we can coerce by an equivalence to obtain the necessary element. Similarly, we can define a map $g : A * (B * C) \to (A * B) * C$, and checking that f and g are inverse to each other is a long and tedious but essentially straightforward computation. $\square$

A more conceptual proof sketch is as follows.

Proof. Let us consider the following diagram where the maps are the obvious projections:

$$\begin{array}{ccccc} A & \longleftarrow & A \times C & \longrightarrow & A \times C \\ \uparrow & & \uparrow & & \uparrow \\ A \times B & \longleftarrow & A \times B \times C & \longrightarrow & A \times C \\ \downarrow & & \downarrow & & \downarrow \\ B & \longleftarrow & B \times C & \longrightarrow & C \end{array}$$

Taking the colimit of the columns gives the following diagram, whose colimit is $(A * B) * C$:

$$A * B \longleftarrow (A * B) \times C \longrightarrow C$$

On the other hand, taking the colimit of the lines gives a diagram whose colimit is $A * (B * C)$.

Hence using a Fubini-like theorem for colimits (that we haven't proved) we have an equivalence $(A * B) * C \simeq A * (B * C)$. The proof of this Fubini theorem for colimits still requires the long and tedious computation, though. □

Lemma 0.1.9. *For any type A, there is an equivalence $\Sigma A \simeq \mathbf{2} * A$.*

Proof. It is easy to define the two maps back and forth and to prove that they are inverse to each other. The details are left as an exercise to the reader. □

We can now construct the Hopf fibration:

Theorem 0.1.10. *There is a fibration over $\mathbb{S}^2$ of fiber $\mathbb{S}^1$ and total space $\mathbb{S}^3$.*

Proof. We proved that $\mathbb{S}^1$ has a structure of H-space (cf Lemma 0.1.7) hence by Lemma 0.1.6 there is a fibration over $\mathbb{S}^2$ of fiber $\mathbb{S}^1$ and total space $\mathbb{S}^1 * \mathbb{S}^1$. But by the two previous results and original lemma 6.5.1 we have:

$$\mathbb{S}^1 * \mathbb{S}^1 = (\Sigma \mathbf{2}) * \mathbb{S}^1 = (\mathbf{2} * \mathbf{2}) * \mathbb{S}^1 = \mathbf{2} * (\mathbf{2} * \mathbb{S}^1) = \Sigma(\Sigma \mathbb{S}^1) = \mathbb{S}^3. \qquad \square$$